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HAIx | HUMAN-AI INTERACTION LAB

Accessibility Interfaces and Emerging Markets

ACM India Summer School 2025: AI for Social Good

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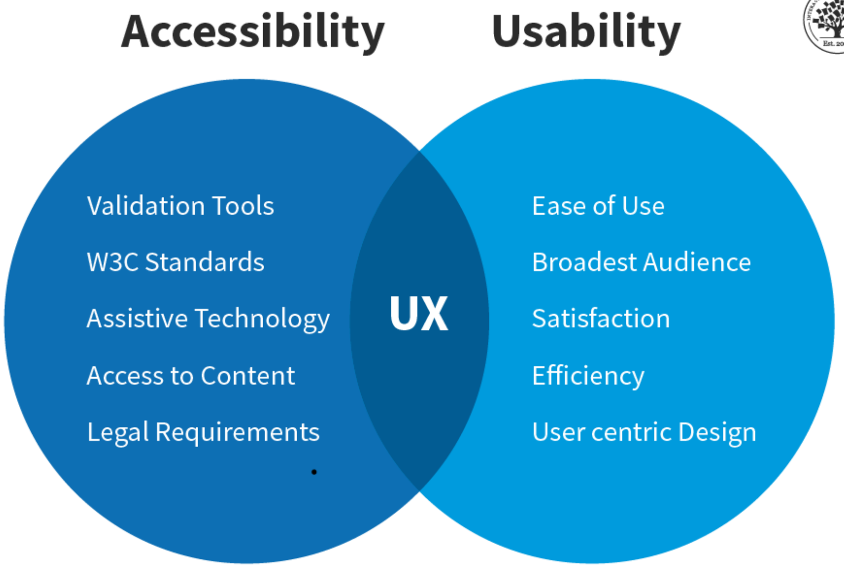
Introduction

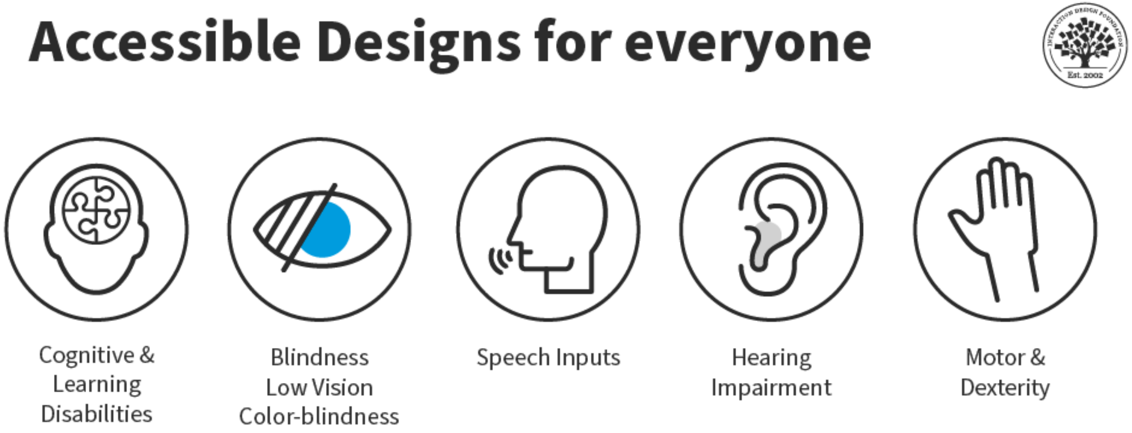
- Emerging technology?
    - Innovations and advancements that are in the early stages of development or adoption and have the potential to significantly impact various aspects of society, industry, or everyday life.

- Accessibility (A11Y) in Computing and HCI?
    - Design and development of systems, software, and interfaces that are usable by all people, including those with disabilities or impairments (e.g., visual, auditory, motor, cognitive)



Accessibility Issues?

- Visual (e.g., color blindness)
- Motor/mobility (e.g., wheelchair-user concerns)
- Auditory (hearing difficulties)
- Seizures (especially photosensitive epilepsy)
- Learning/cognitive (e.g., dyslexia)



Why this matters

- An estimated 1.3 billion people – or 1 in 6 people worldwide – experience significant disability.
- Persons with disabilities die earlier, have poorer health, and experience more limitations in everyday functioning than the rest of the population due to health inequities.
- Accessibility interfaces that enable users with disabilities to interact with systems e.g., screen readers, eye-tracking, brain signals, voice commands, switch controls, haptic feedback etc.

SOTA Assistive Tech for effective communication and mobility

- Augmentative and Alternative Communication (AAC), wheelchair based assistive devices

Mobility Challenges

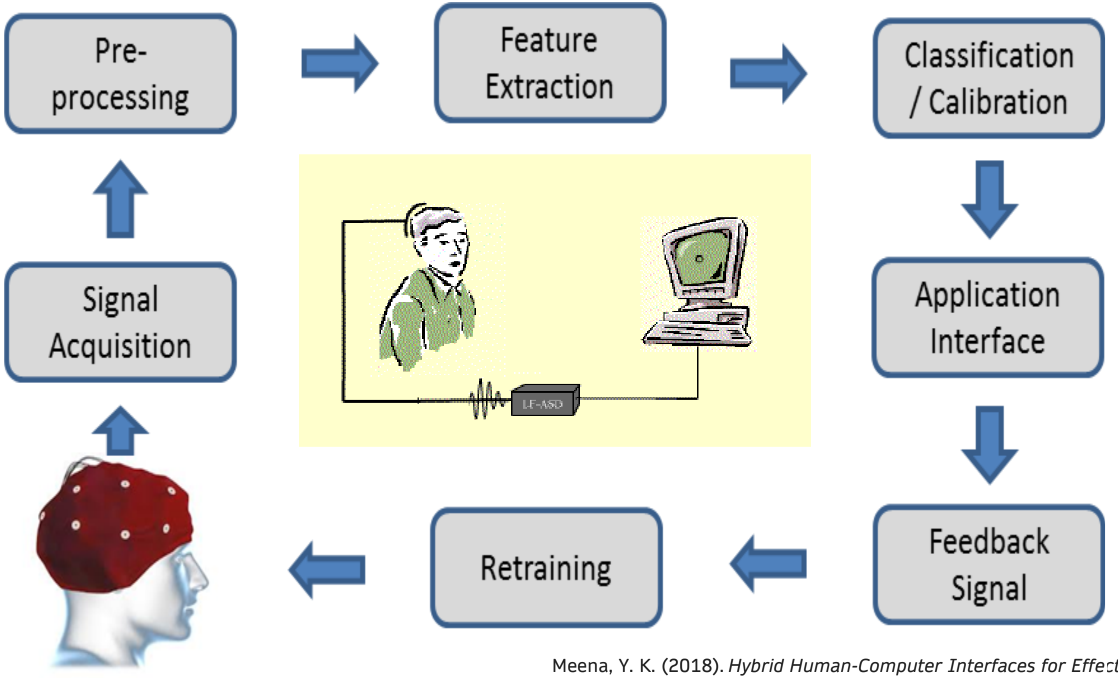
- Physical Impairments
  - Conditions like spinal cord injuries, cerebral palsy, muscular dystrophy, and stroke can severely limit muscle control and movement
- Dependence on Assistive Devices
  - Wheelchairs, walkers, and other assistive devices are often necessary for basic mobility
- Environmental Barriers
  - Physical barriers like stairs, narrow doorways, and inaccessible public spaces can further restrict mobility

Communication Challenges

- Speech Impairments
  - Conditions like aphasia, dysarthria, and vocal cord paralysis can affect speech clarity and comprehension
- Limited Vocabulary
  - Difficulty in finding the right words or forming complete sentences can hinder communication
- Non-Verbal Communication
  - Reliance on gestures, facial expressions, and body language can be difficult to interpret

Technologies and Accessibility Interfaces

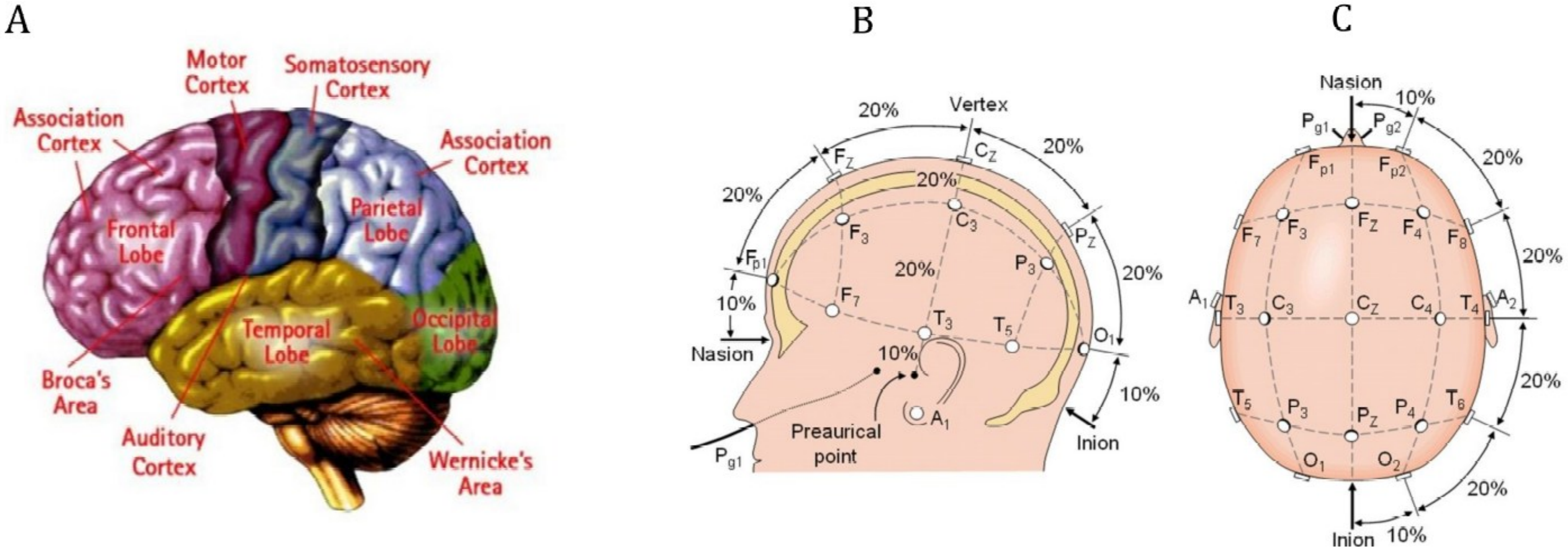
Brain-computer interface (BCI)



Meena, Y. K. (2018). *Hybrid Human-Computer Interfaces for Effective Communication and Independent Living* (Doctoral dissertation, Ulster University).

Users →Totally locked-in patients →Quadriplegia →Healthy people

Overview of the brain



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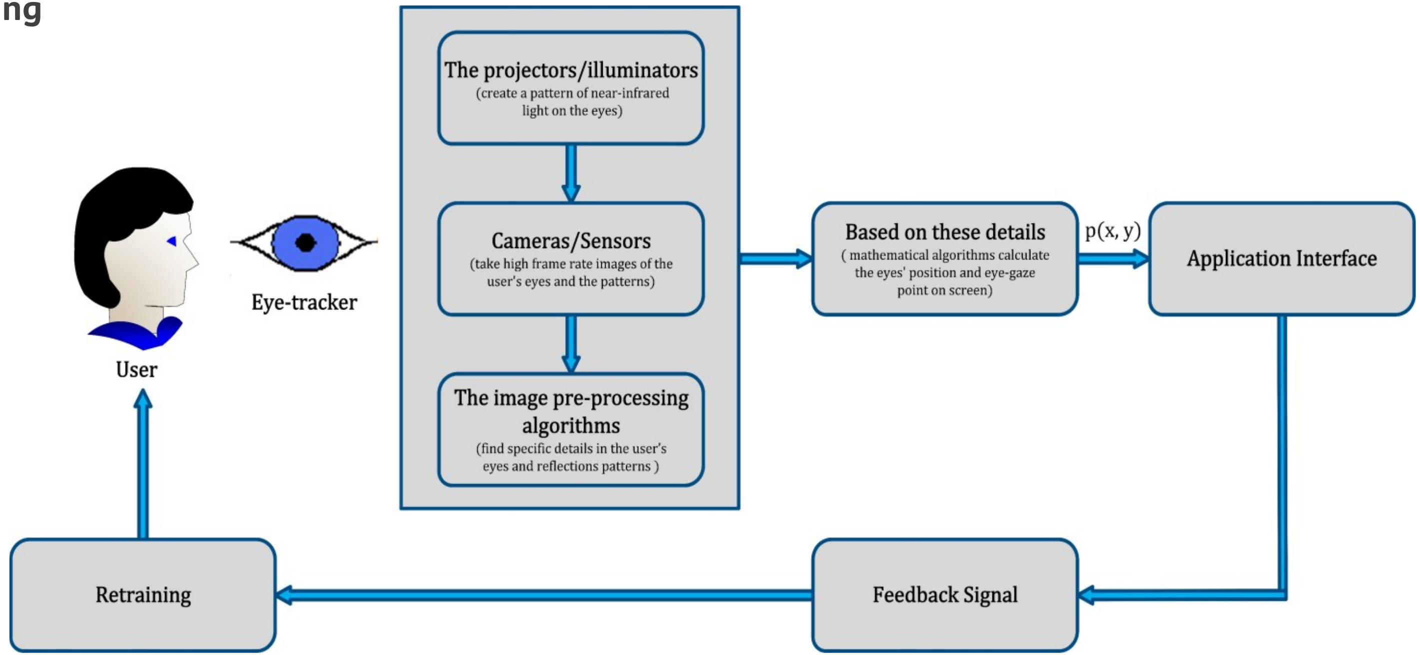
Technologies and Accessibility Interfaces

Eye-tracking



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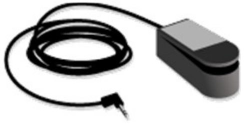
Technologies and Accessibility Interfaces

Single-switch hardware interfaces

Cuddly switch

USB click switch

Specs switch



Micro light switch

Plate switch

Sip/puff switch

Pedal switch

Foot switch

Inclusive simple switch box

USB switch interface plus

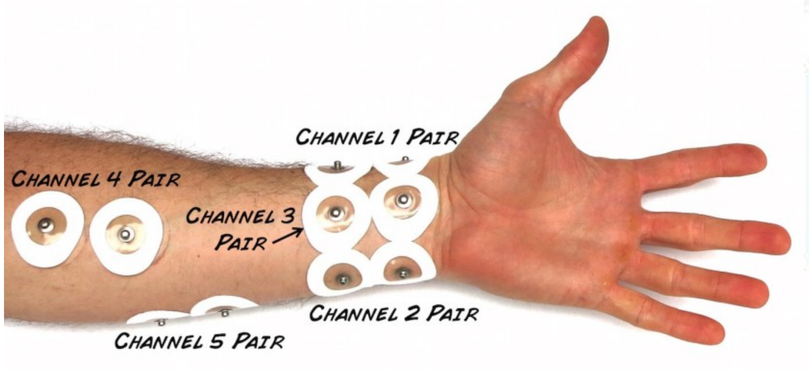
Joy Cable 2

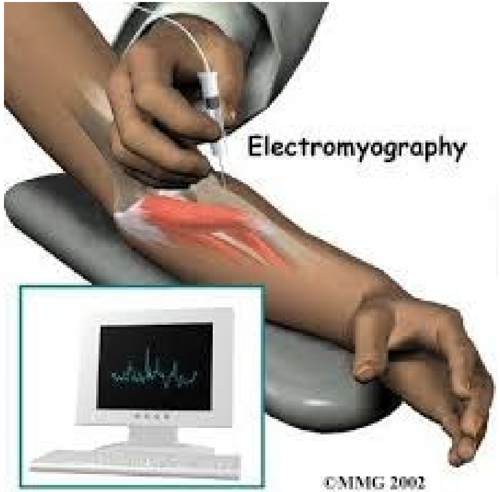
Crick USB Switch Box

Technologies and Accessibility Interfaces

Electromyography

- Surface EMG (sEMG) and
- Intramuscular EMG





Challenges with technologies and interfaces

- Efficient implementation of the physical access switches and EMG gestures, some amount of motor activity is required by the user to control the device.

Eye-tracking Limitations

- Midas Touch Problem
- Dwell-time issue
- Frequent recalibration
- Accuracy of the detection of the gaze coordinates

BCI Limitations

- Inconvenient setup procedures
- Lack of sufficiently high accuracy
- Reliability
- Low information transfer rate (ITR)
- User acceptability

Gaze and Motor Imagery based Multimodal/Hybrid BCI System

Comprehensive design of a hybrid BCI system

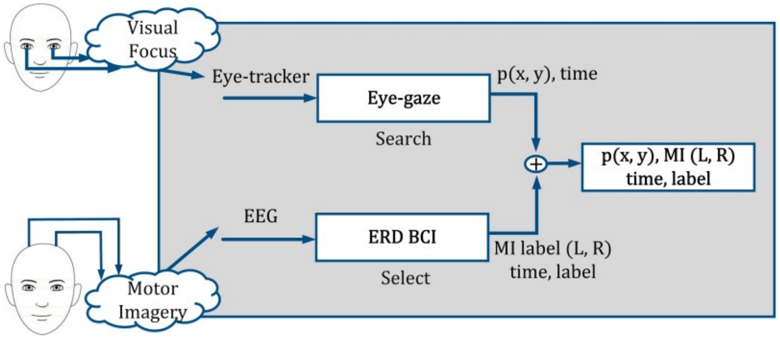
- Commands /choices in BCI and eye-tracking system
- False positives selection of eye-tracking system
- Designing a natural user interface
- Frequent recalibration of eye-tracking system

Developed model, system, and approaches

- Hybrid BCI that combines an ERD BCI and an eye tracker
- Gaze Detection
- Feature Extraction and classification

Gaze and Motor Imagery based Hybrid System

Proposed Design Model of the Hybrid BCI System



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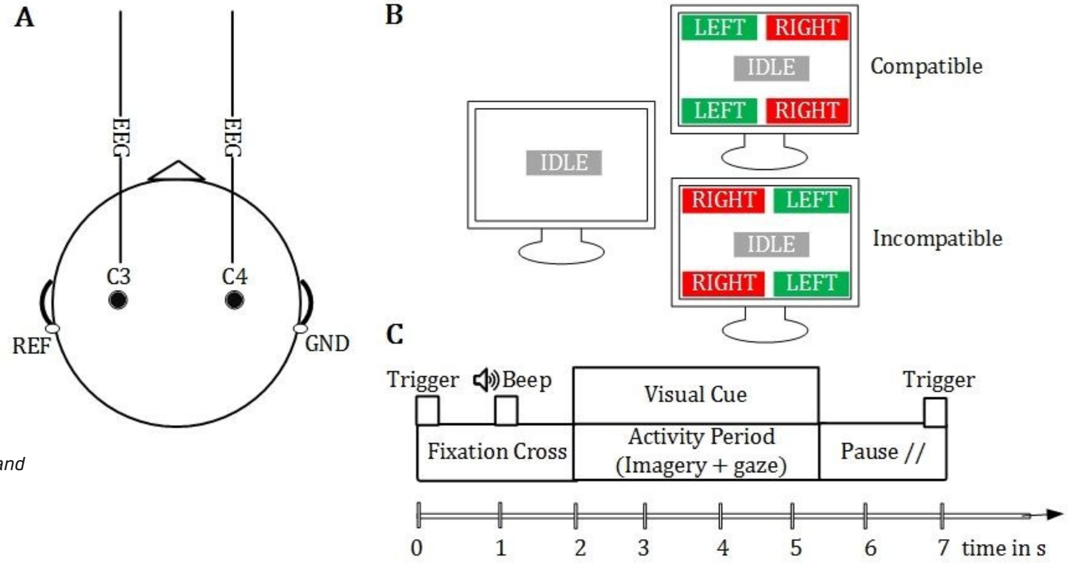

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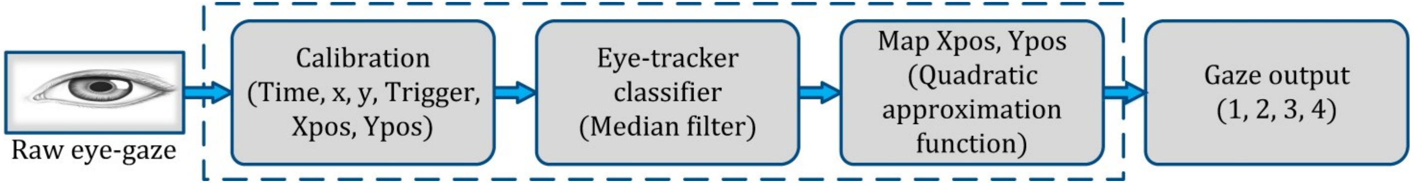
Meena, Y. K et al., (2015). Simultaneous gaze and motor imagery hybrid BCI increases single-trial detection performance: a compatible incompatible study. In Proceedings of the IEEE-EMBS.

Gaze and Motor Imagery based Hybrid System

Signal Processing and Classification

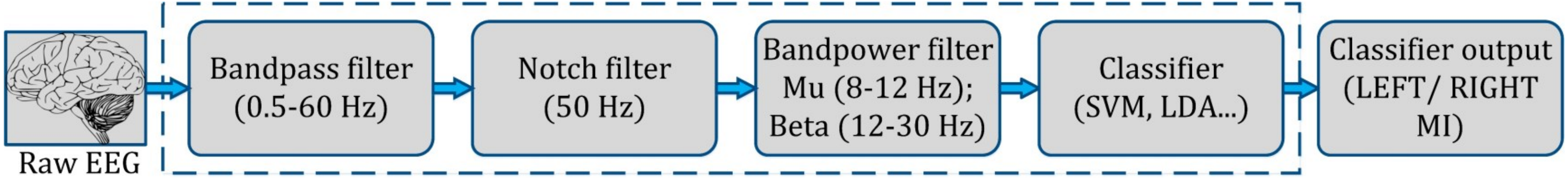
Eye-gaze detection of four classes: Confusion matrix



EEG Feature Extraction

- Temporal filtering: Band power feature – mu (8-12 Hz) and beta (12-30 Hz)
- Spatial filtering: Common spatial patterns (CSP)

EEG Classification: SVM, LDA, kNN

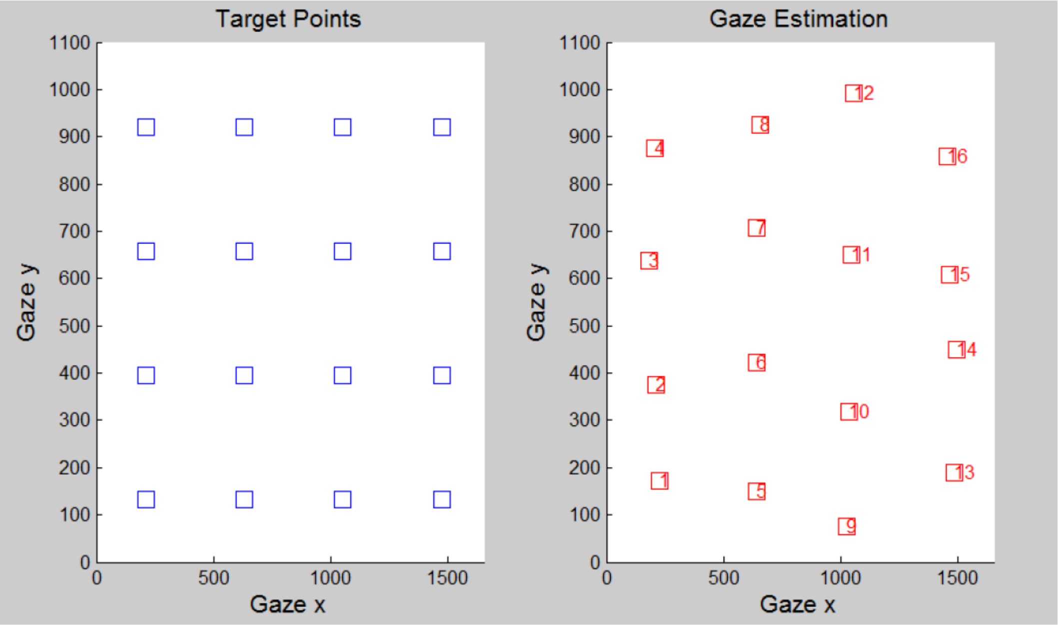


Information transfer rate (ITR): bits per symbol

$$ITR_{Wolpaw} = log_2 N + B.log_2 B + (1 - B).log_2[(1 - B)/(N - 1)]$$

Gaze and Motor Imagery based Hybrid System

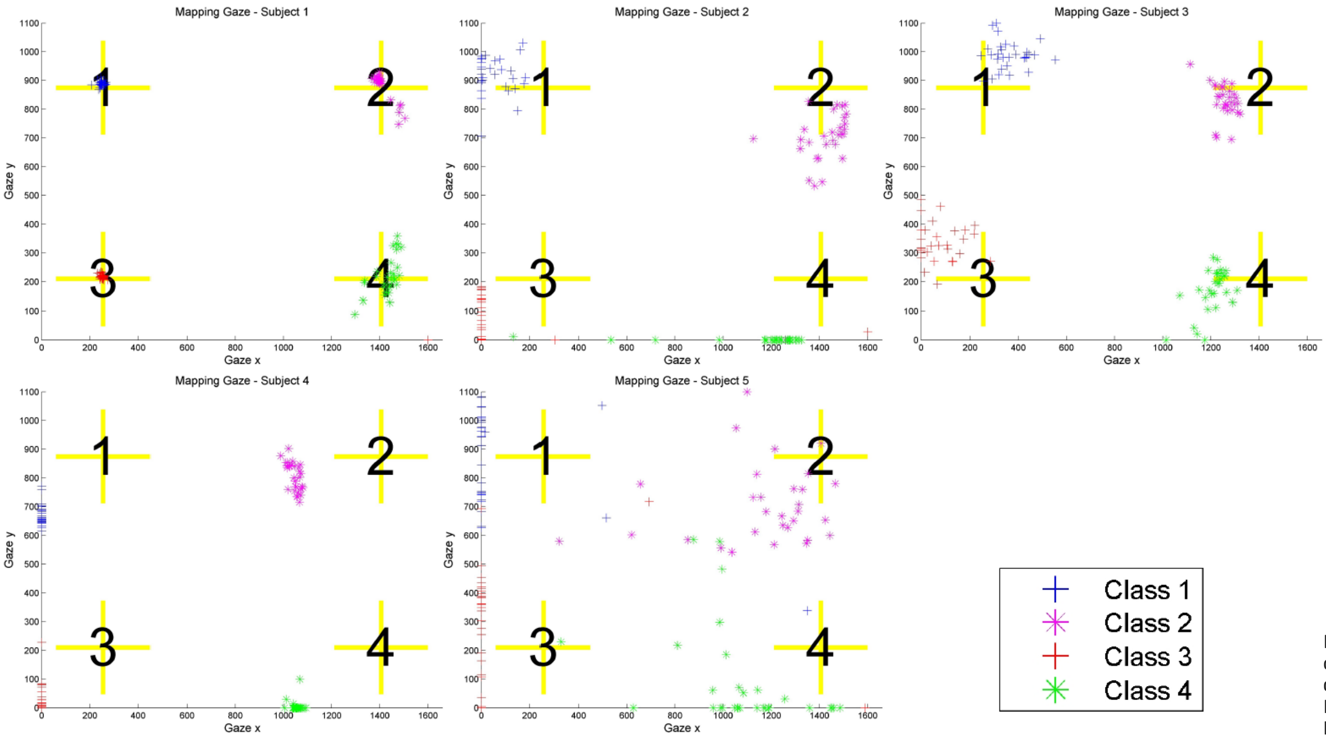
Eye-gaze estimation results for a single calibration test run

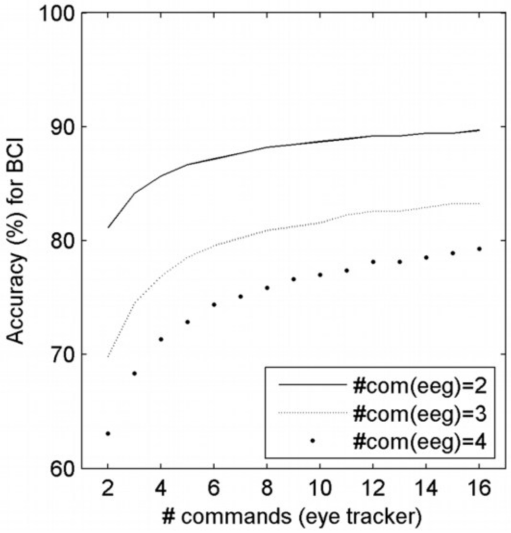


Gaze and Motor Imagery based Hybrid System

Results

- BCI performance increase with higher number of commands





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Adaptive Multimodal Virtual Keyboard for Alternative and Augmentative Communication

Multimodal accessibility and user-based adaptation

- Affordability, portability, and practicality
- Adaptation to the user
- Usability and accuracy of gaze based virtual keyboards
- Accuracy of the gaze coordinates
- Frequent recalibration of eye-tracking system

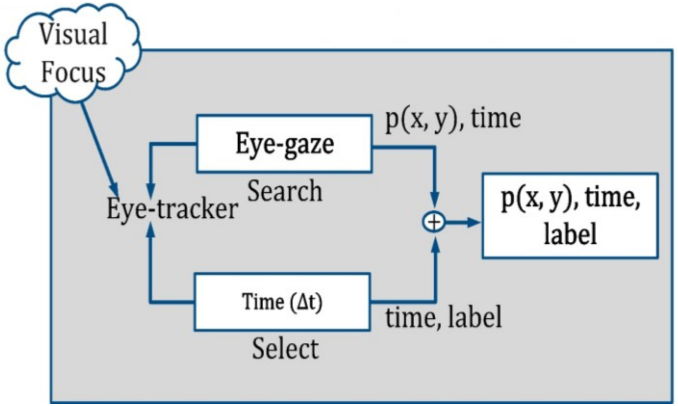
Developed methods and approaches

- Non-adaptive methods
- Adaptive methods
- Dwell-free mechanisms

Adaptive Multimodal Virtual Keyboard for Alternative and Augmentative Communication

Gaze-based access control

- Asynchronous mode
- Synchronous mode
- The total number of commands that are available at any time in the system by M.
- Each command C_i is defined by the coordinates corresponding to the center of its box (x_c^i, y_c^i), i $\in$ {1 … M}
- The gaze coordinates at time t by (x_t, y_t) then the distance between a command box and the current gaze position, d_t^i is defined by its Euclidean distance as: $\boldsymbol{d_t^i} = \sqrt{(\boldsymbol{x_c^i} - \boldsymbol{x_t})^2 + (\boldsymbol{y_c^i} - \boldsymbol{y_t})^2}$



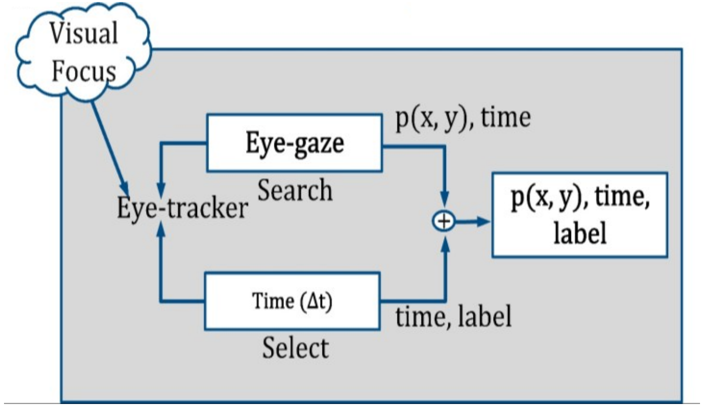
Adaptive Multimodal Virtual Keyboard for Alternative and Augmentative Communication

Gaze-based access control

- Asynchronous mode

Algorithm 1 Command selection - asynchronous mode

$t \leftarrow 0, \delta \leftarrow 0$
$\text{select}_0 \leftarrow -1$
while (true) **do**
  $\text{select}_t \leftarrow \text{argmin}_{1 \le i \le M}(d_t^i)$
  if $(\text{select}_t == \text{select}_{t-1})$ **then**
    $\delta \leftarrow \delta + 1$
    $v \leftarrow 255 * (\Delta t_0 - \delta)/\Delta t_0$ {update color}
  else
    $\delta \leftarrow 0$
  if $(\delta \geq \Delta t_0)$ **then**
    Run command (select_t)
    $\delta \leftarrow 0$
  $t \leftarrow t + 1$



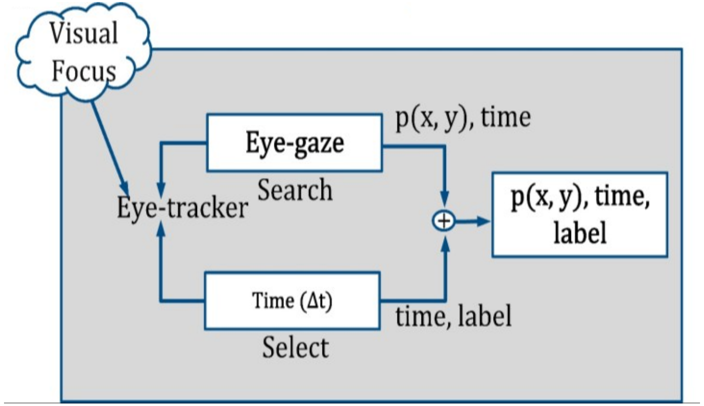
Adaptive Multimodal Virtual Keyboard for Alternative and Augmentative Communication

Gaze-based access control

- Synchronous mode

Algorithm 2 Command selection - Synchronous mode

$\alpha_1 \leftarrow 0.5$
while (true) **do**
  $w(i) \leftarrow 0, \forall i \in \{1..M\}$
  for $t \leftarrow 1$ **to** Δt_1 **do**
    $\text{select}_t \leftarrow \text{argmin}_{1 \leq i \leq M}(d_t^i)$
    $w(\text{select}_t) \leftarrow w(\text{select}_t) + \sqrt{t}$
  $\text{select}_s \leftarrow \text{argmax}_{1 \leq i \leq M}(w)$
  $P(\text{select}_s) \leftarrow \frac{max(w)}{\sum_{j=1}^{M} w(j)}$
  if $(P(\text{select}_s) \geq \alpha_1)$ **then**
    Run command (select_s)
  else
    $\text{select}_s \leftarrow -1$



Adaptive Multimodal Virtual Keyboard for Alternative and Augmentative Communication

Gaze-based access control

- Adaptive dwell time in asynchronous mode

Algorithm 3 Adaptive dwell time

$\beta_1 \equiv 500\ ms$, $\epsilon_1 \equiv 500\ ms$, $\epsilon_2 \leftarrow 0.5$
if $(N_{cor}/N_h > \epsilon_2)$ **then** {rule #1}
    $\Delta t_0 \leftarrow \Delta t_0 + \beta_1$
if $(|\overline{\Delta t_c}(k) - \Delta t_0| \leq \epsilon_1)$ **then** {rule #2}
    $\Delta t_0 \leftarrow \Delta t_0 - \beta_1$

The current average of Δtc over the past Nh commands

$$\overline{\Delta t_c}(k) = \frac{1}{N_h} \sum_{k_0=1}^{N_h} \Delta t_c(k - k_0)$$

- Adaptive trial period in synchronous mode

Algorithm 4 Adaptive trial period

$\beta_2 \equiv 500\ ms$, $\epsilon_2 \leftarrow 0.5$, $\epsilon_3 \leftarrow 0.9$
if $(\overline{P(\text{select}_s)_k} > \epsilon_3)$ **then** {rule #1}
    $\Delta t_1 \leftarrow \Delta t_1 - \beta_2$
if $(N_r/N_h \geq \epsilon_2)$ **then** {rule #2}
    $\Delta t_1 \leftarrow \Delta t_1 + \beta_2$
if $(N_{cor}/N_h \geq \epsilon_2)$ **then** {rule #3}
    $\Delta t_1 \leftarrow \Delta t_1 + \beta_2$

The average probability to detect a command in the kth trial b
considering the last Nh previous trials

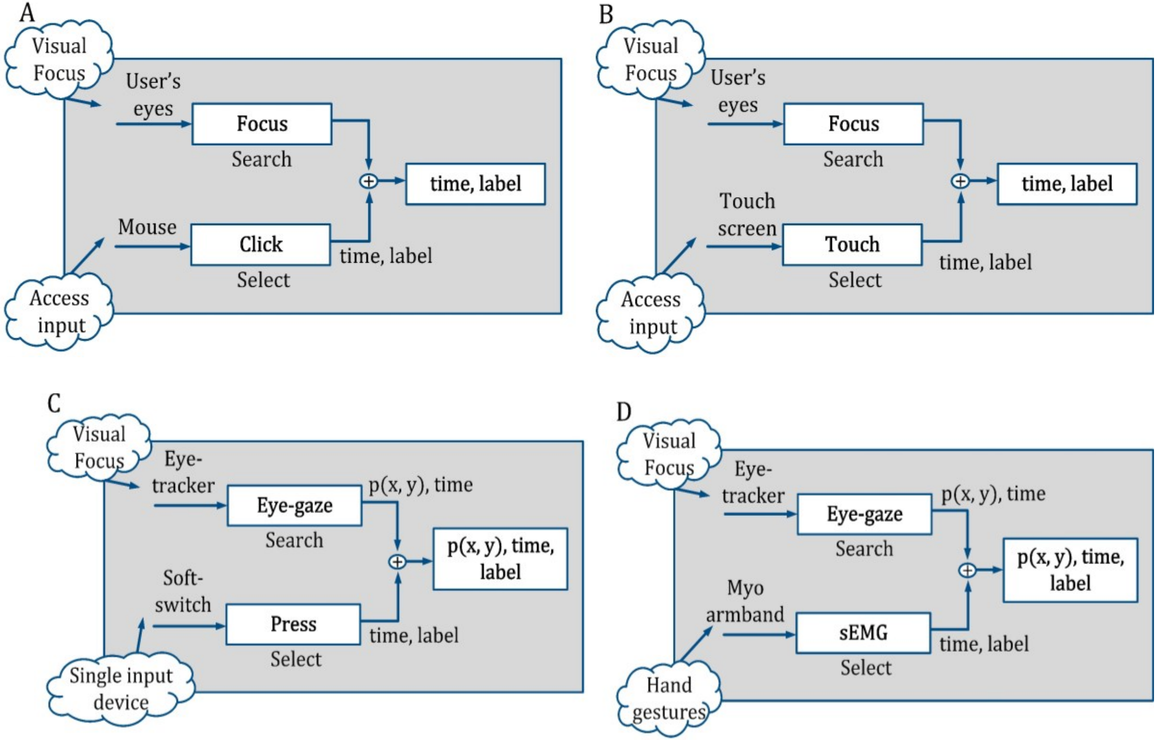
$$\overline{P(\text{select}_s)_k} = \frac{1}{N_h} \sum_{k_0=1}^{N_h} P(\text{select}_s)_{k-k_0}$$

Adaptive Multimodal Virtual Keyboard for Alternative and Augmentative Communication

Dwell-free mechanisms

- Command selection with single modality

- Command selection with multimodality



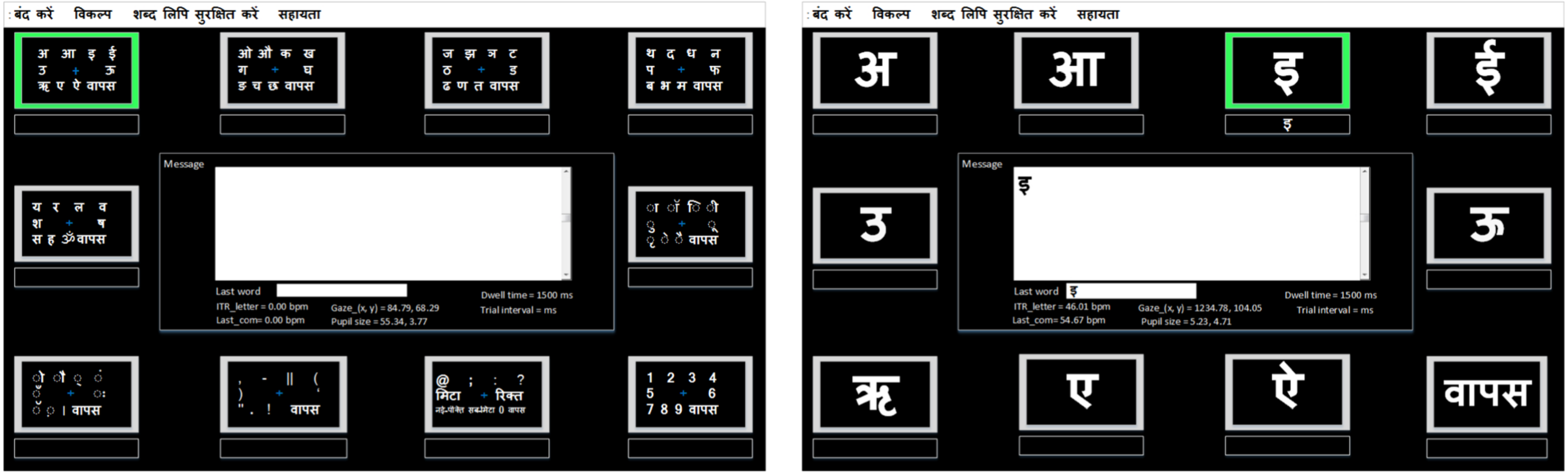


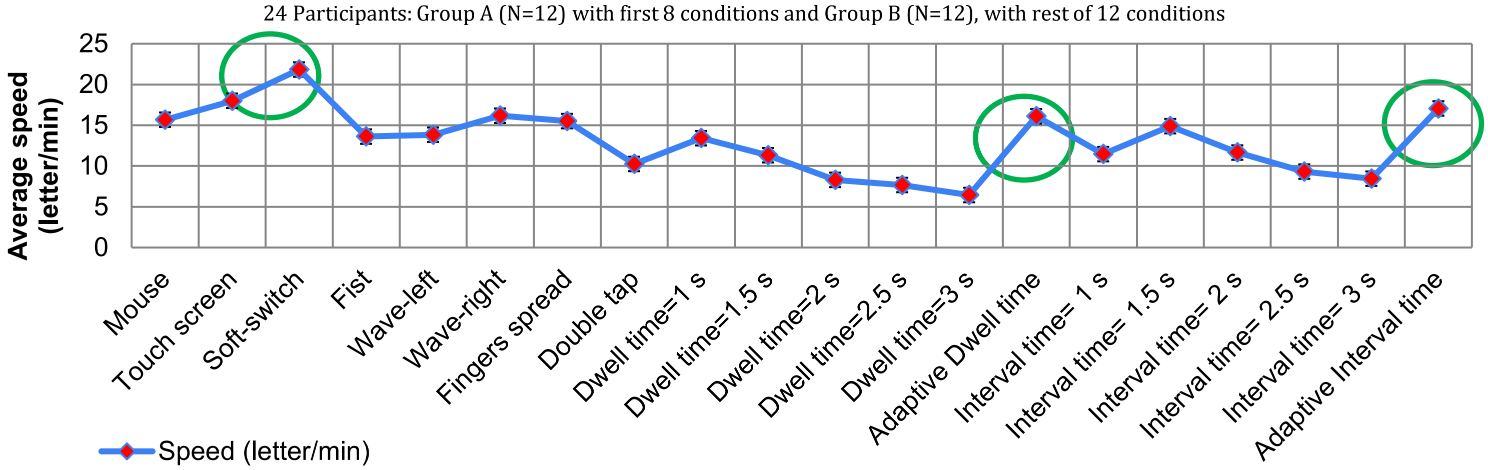
Figure 1: Layout of proposed Hindi virtual keyboard application in level one when c1 is selected (left) and level two after the selection of c1 (right), with the ten commands (from left to right, top to bottom).

[1] Meena, Yogesh Kumar, Hubert Cecotti, KongFatt Wong-Lin, and Girijesh Prasad. "Design and evaluation of a time adaptive multimodal virtual keyboard." *Journal on Multimodal User Interfaces 13 (2019): 343-361*

Adaptive Multimodal Virtual Keyboard for Alternative and Augmentative Communication

Results

- Experiment 1: mouse versus touch screen
- Experiment 2: eye-tracker with soft-switch versus eye-tracker with sEMG based hand gestures
- Experiment 3: fixed versus adaptive dwell time with eye-tracker asynchronous mode
- Experiment 4: fixed versus adaptive trial period with eye-tracker in synchronous mode



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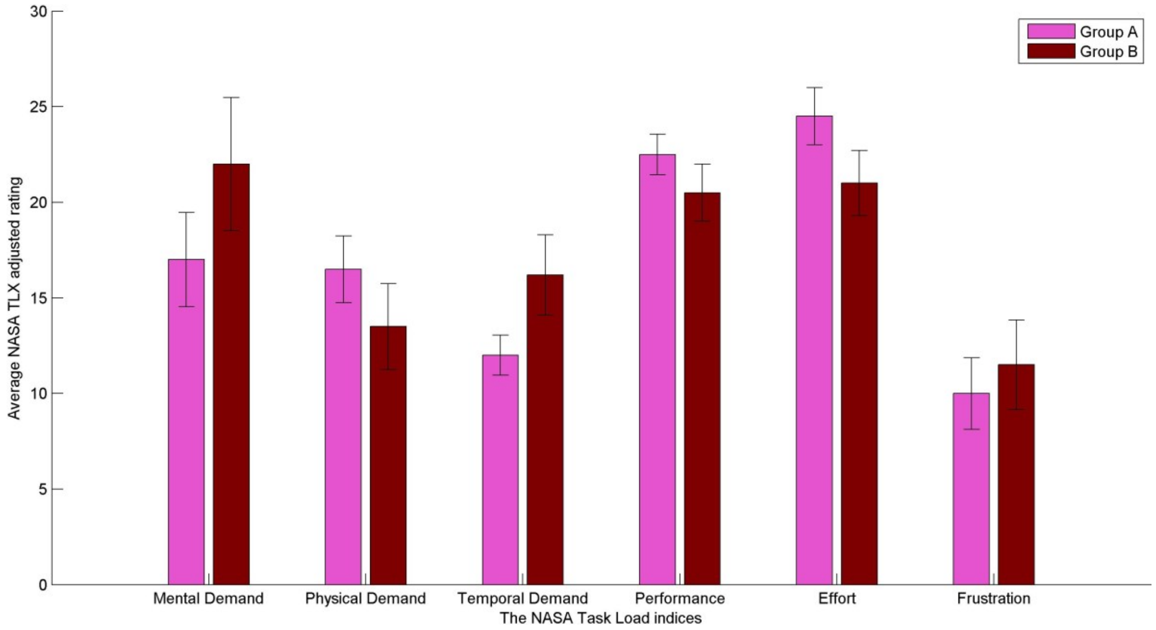
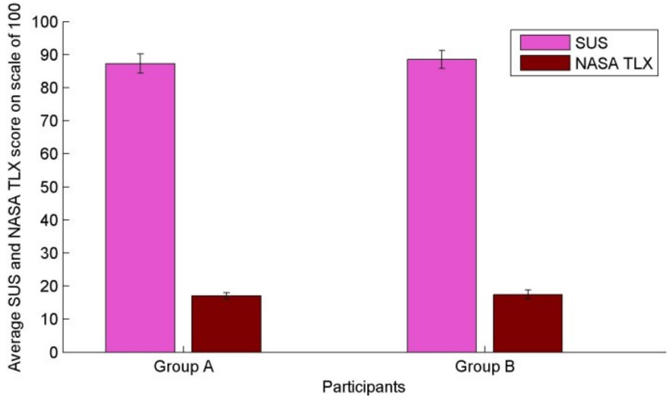
Adaptive Multimodal Virtual Keyboard for Alternative and Augmentative Communication

Results

- Subjective evaluation





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Toward Optimization of Gaze-Controlled Human–Computer Interaction: Application to Hindi Virtual Keyboard for Stroke Patients[2]

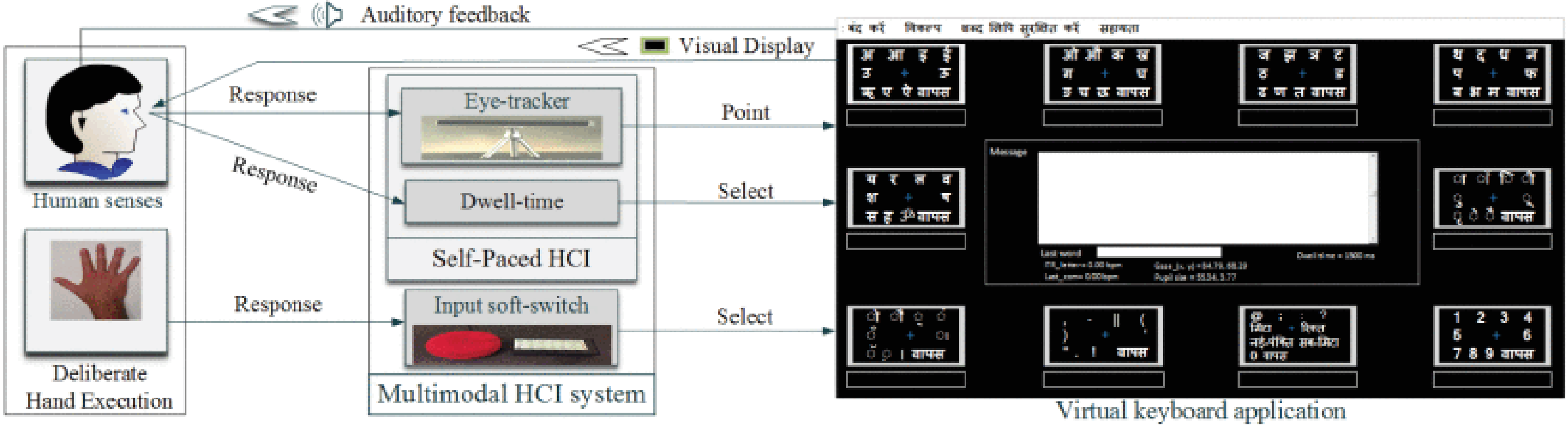

Figure 4: The main components of the proposed multimodal human computer interaction (HCI) system wherein the eye-tracker only and eye-tracker with a single input switch are used for searching and selection of the items on virtual keyboard application.

[2] Meena, Yogesh Kumar, Hubert Cecotti, Braj Bhushan, Ashish Dutta, and Girijesh Prasad. "Detection of Dyslexic Children Using Machine Learning and Multimodal Hindi Language Eye-Gaze-Assisted Learning System." *IEEE Transactions on Human-Machine Systems 53, no. 1 (2023): 122-131.*

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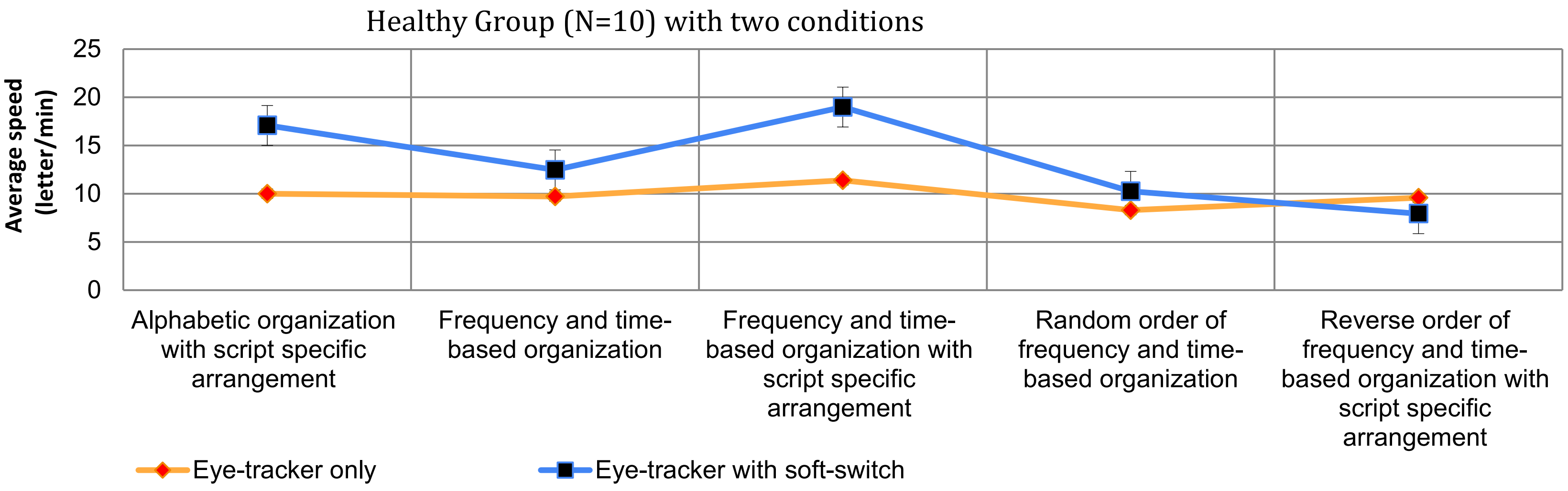

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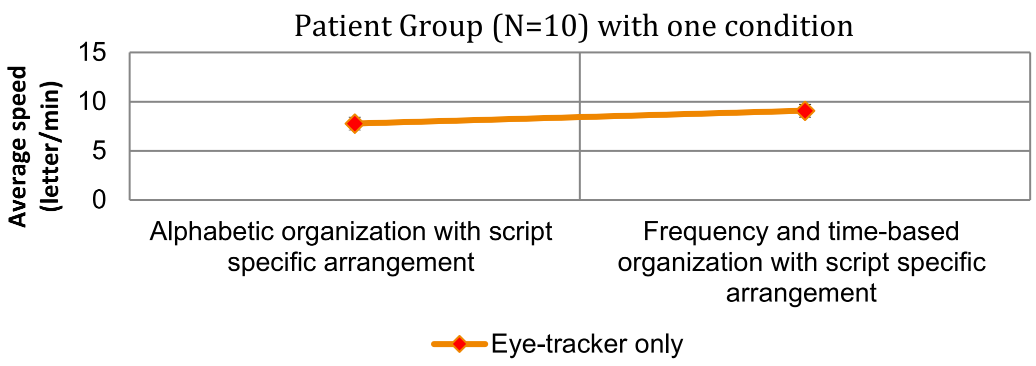


2 Meena, Yogesh Kumar, Hubert Cecotti, Braj Bhushan, Ashish Dutta, and Girijesh Prasad. "Detection of Dyslexic Children Using Machine Learning and Multimodal Hindi Language Eye-Gaze-Assisted Learning System." *IEEE Transactions on Human-Machine Systems 53, no. 1 (2023): 122-131.*

Detection of Dyslexic Children Using Machine Learning and Multimodal Hindi Language Eye-Gaze-Assisted Learning System[3]

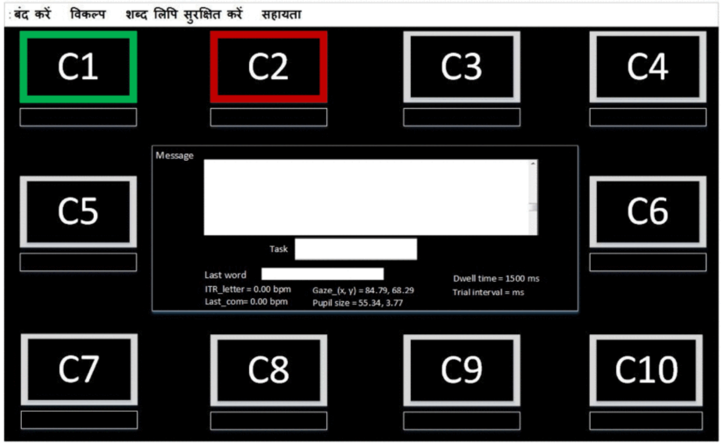


Figure 5: Positions of ten commands (c1)–(c10) in the Hindi language VK. Two types of visual positive (for intended item selection) and negative feedback (for accidental item selection) are shown in commands C1 and C2, respectively.

Index	Hindi word	# letters	# diacritics	# commands
1	घर	2	0	4
2	नमक	3	0	6
3	बचपन	4	0	8
4	जलमहल	5	0	10
5	इधरउधर	6	0	12
6	राम	3	1	6
7	इमली	4	1	8
8	तरबूज	5	1	10
9	किसतरह	6	1	12
10	संगमरमर	7	1	14
Total	-	38	5	76

Figure 6: Predefined words (i.e., copy spelling) task. The maximum duration given to the participants for the first five words are 40, 60, 80, 100, and 120 s, respectively; it was 60, 80, 100, 120, and 140 s, respectively, for the last five words. The maximum duration was chosen as 20 s per character (letter or matra).

3 Meena, Yogesh Kumar, Hubert Cecotti, Braj Bhushan, Ashish Dutta, and Girijesh Prasad. "Detection of Dyslexic Children Using Machine Learning and Multimodal Hindi Language Eye-Gaze-Assisted Learning System." *IEEE Transactions on Human-Machine Systems 53, no. 1 (2023): 122-131.*

Detection of Dyslexic Children Using Machine Learning and Multimodal Hindi Language Eye-Gaze-Assisted Learning System[3]

TABLE VI
PERCENTILE RANK AND DESCRIPTIVE CLASSIFICATION OF VCI INDEX WITH TYPING PERFORMANCE FOR CHILDREN WITH DYSLEXIA

VCI Percentile Rank	Descriptive Classification	Group Sample	Speed (letter/min) TS Mean		ET Mean		ETSS Mean	
			Mean	Std.	Mean	Std.	Mean	Std.
9-23	Low Average	n=8	10.61	1.25	7.42	0.48	12.98	1.43
2-8	Borderline	n=5	9.21	1.03	7.33	0.33	10.87	1.39
0.01-2	Intellectual Disability	n=3	7.28	1.32	6.89	0.61	9.49	1.57

TABLE VII
PERCENTILE RANK AND DESCRIPTIVE CLASSIFICATION OF PRI INDEX WITH TYPING PERFORMANCE FOR CHILDREN WITH DYSLEXIA

PRI Percentile Rank	Descriptive Classification	Group Sample	Speed (letter/min) TS		ET		ETSS	
			Mean	Std.	Mean	Std.	Mean	Std.
25-73	Average	n=6	10.96	2.70	8.23	1.47	12.21	2.58
9-23	Low Average	n=3	10.06	1.46	7.17	0.75	11.92	2.13
2-8	Borderline	n=3	7.43	1.15	6.48	0.09	9.72	1.16
0.01-2	Intellectual Disability	n=4	7.66	0.75	5.57	0.89	10.78	1.14

[3] Meena, Yogesh Kumar, Hubert Cecotti, Braj Bhushan, Ashish Dutta, and Girijesh Prasad. "Detection of Dyslexic Children Using Machine Learning and Multimodal Hindi Language Eye-Gaze-Assisted Learning System." *IEEE Transactions on Human-Machine Systems 53, no. 1 (2023): 122-131.*

Detection of Dyslexic Children Using Machine Learning and Multimodal Hindi Language Eye-Gaze-Assisted Learning System[3]

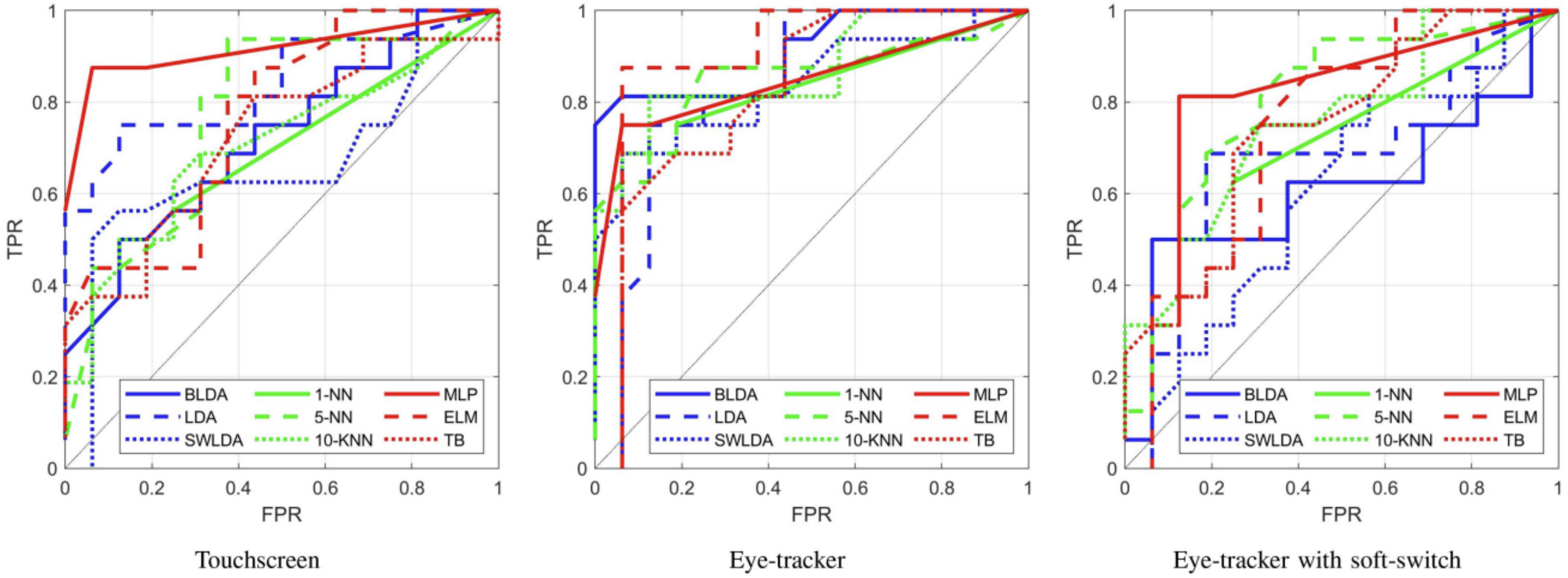
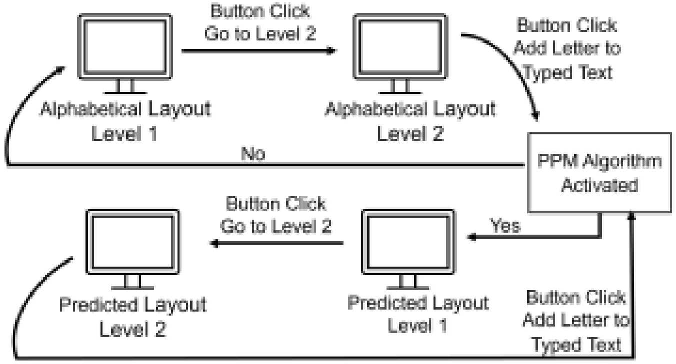


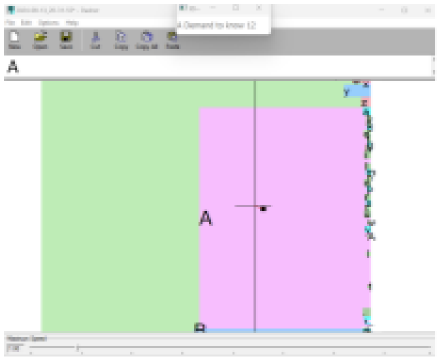
Fig. 6. ROC curve for each condition for the detection of dyslexic children based on the typing speed.

3 Meena, Yogesh Kumar, Hubert Cecotti, Braj Bhushan, Ashish Dutta, and Girijesh Prasad. "Detection of Dyslexic Children Using Machine Learning and Multimodal Hindi Language Eye-Gaze-Assisted Learning System." *IEEE Transactions on Human-Machine Systems 53, no. 1 (2023): 122-131.*

Flex-Tree: Gaze-Controlled Interface with Context-Based Letter and Word Predictions[4]

Develop the Prediction by Partial Matching method for a tree-based virtual keyboard interface





4 Etikikota Hrushikesh, Meena, Yogesh Kumar. "Predictive Tree-based Virtual Keyboard for Improved Gaze Typing." *IEEE SMC 2024.*

Flex-Tree: Gaze-Controlled Interface with Context-Based Letter and Word Predictions[4]

Condition		Mouse						Eye-tracker					
		HandFT			RandFT			HandFT			RandFT		
		Speed	ITR_*com*	ITR_*letter*	Speed	ITR_*com*	ITR_*letter*	Speed	ITR_*com*	ITR_*letter*	Speed	ITR_*com*	ITR_*letter*
NoPPM	Mean	21.7	143.9	133.7	24.1	160.5	149.1	13.3	88.4	82.1	13.8	91.8	85.3
	Std	07.3	48.4	44.9	04.4	29.5	27.4	02.9	19.5	18.1	02.1	13.9	12.9
PPM1	Mean	23.7	157.4	146.3	26.4	175.5	163	15.3	102.1	94.9	15.2	101.1	94.0
	Std	04.9	32.9	30.6	06.8	45.1	41.9	02.6	17.6	16.4	01.8	12.4	11.5
PPM2	Mean	30.3	201.6	187.3	28.3	188.1	174.8	15.7	104.6	97.2	15.6	103.5	96.2
	Std	09.1	60.9	56.6	07.4	49.1	45.7	03.4	22.6	21.0	02.8	18.8	17.4
PPM3	Mean	27.5	182.9	169.9	27.7	184.3	171.2	16.0	106.5	99.0	16.3	108.4	100.7
	Std	07.7	51.3	47.7	08.9	59.4	55.2	02.6	17.8	16.5	02.9	19.7	18.3

Condition		Dasher Typing Performance				Deletion Time	
		Mouse		Eye-tracker		Mouse	
		Speed	ITR_*letter*	Speed	ITR_*letter*	Flex-Tree	Dasher
HandDS	Mean	09.1	56.6	06.8	42.1	01.4	04.2
	Std	03.0	18.6	03.2	20.3	00.1	01.0
RandDS	Mean	10.3	63.8	04.9	30.5	01.4	04.5
	Std	03.3	20.3	01.7	10.5	0.16	01.0

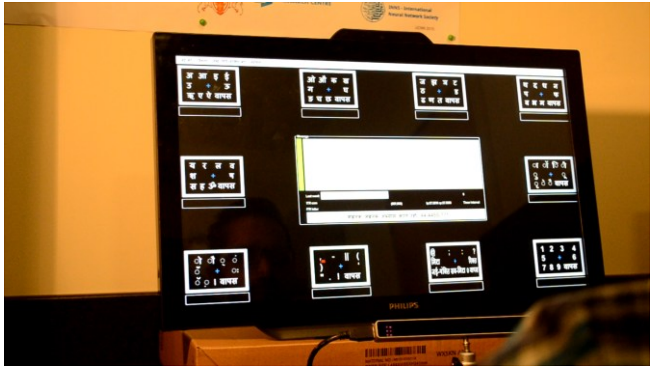
Algorithm 1 PPM for Tree-Based Interfaces

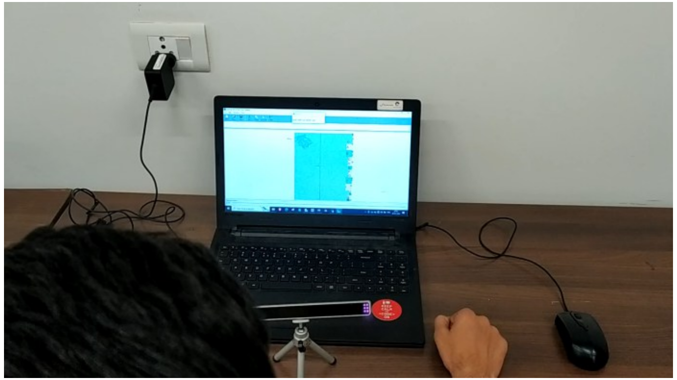
```
Input:Change_level_to-Indicates the next level, either 1 or 2.
Input:K-Degree of PPM used or length of context
Input:command_id- ID of Button just clicked in either level
Output:Candidates- A list of strings representing the text in
  each button command
if Change_level_to = 1 then
    Add the character on command button to the text_entered
    if text_entered < k then
        candidates ← group all the characters alphabetically
    else
        sets1 ← Most Probable Characters are grouped.
        sets2 ← Most frequent letters in language are grouped
        sets3 ← Remaining characters are grouped
        add set1, set2, and set3 to candidates
    end if
else
    set1 ← 8 characters from the command_id in level 1
    add set1 to candidates
    add 'BACK' at fifth position of the candidates
end if
add 'DELETE' at sixth position of the candidates
return Candidates
```

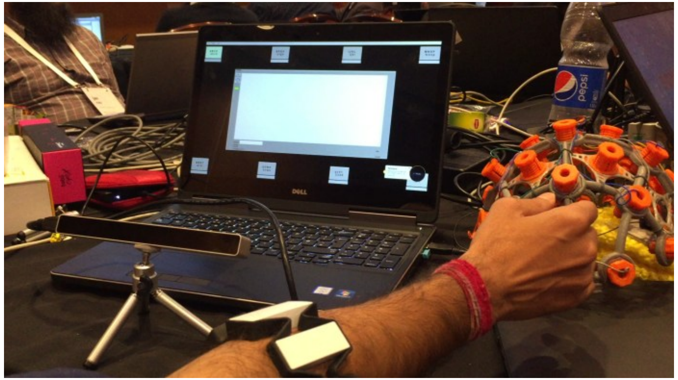
Note: Grouping in the algorithm always indicates the formation of strings of length 8 characters.

4 Etikikota Hrushikesh, Meena, Yogesh Kumar. "Predictive Tree-based Virtual Keyboard for Improved Gaze Typing." *IEEE SMC 2024.*

Working prototype demos





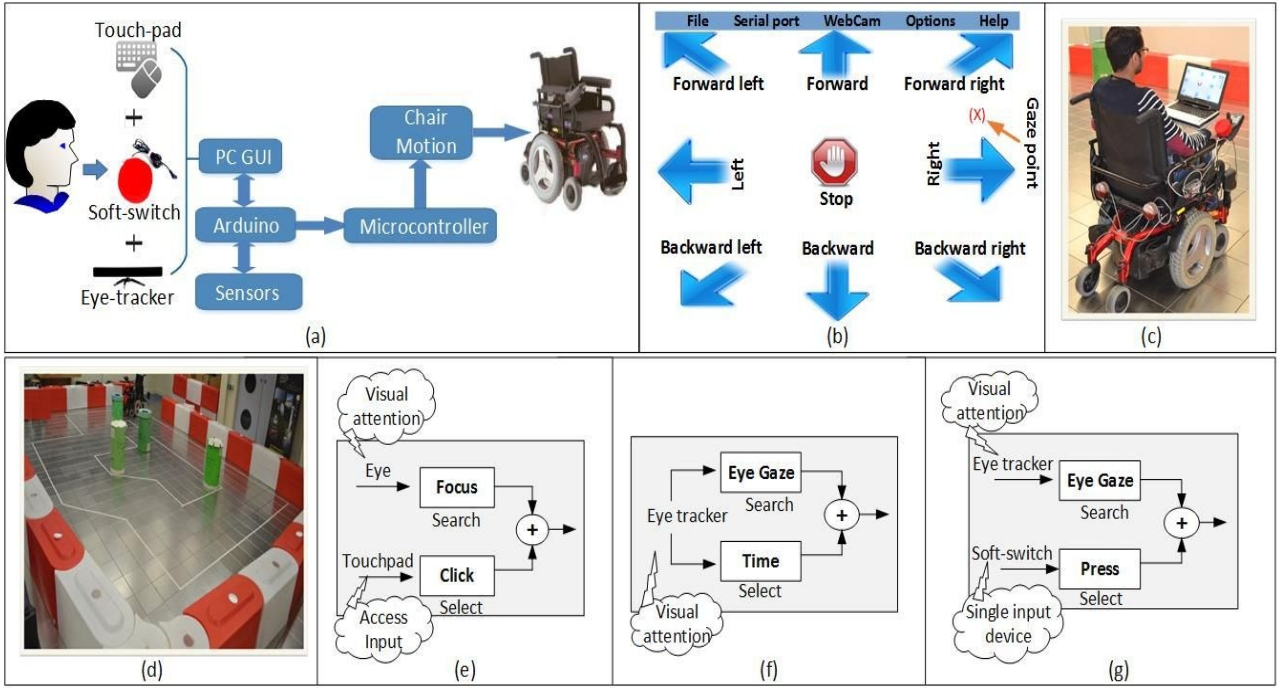


Currently available in 3 languages: English, Hindi, Bangla

Assistive Technology for Independent Living: Smart Multimodal Wheelchair Control

Implementation of HCI for wheelchair control and related applications

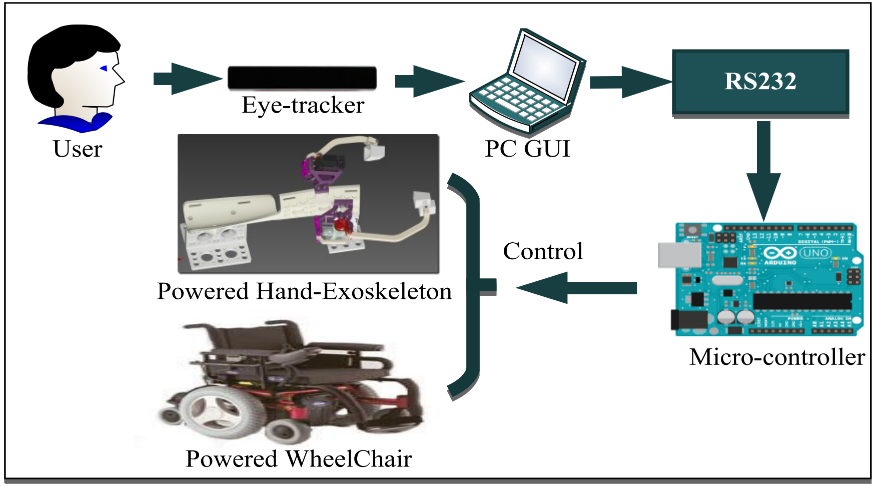
- Adaptation user interface
- Multimodal facilities
- Portability
- Integration
- Efficient user interface

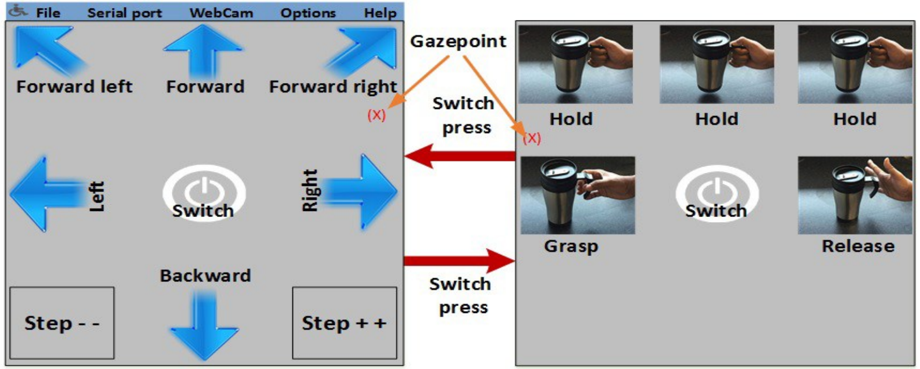


Meena, Y. K., et al., (2018). A multimodal interface to resolve the Midas-Touch problem in gaze controlled wheelchair. In Proceedings of the IEEE-EMBC.

Assistive Technology for Independent Living: Smart Multimodal Wheelchair Control

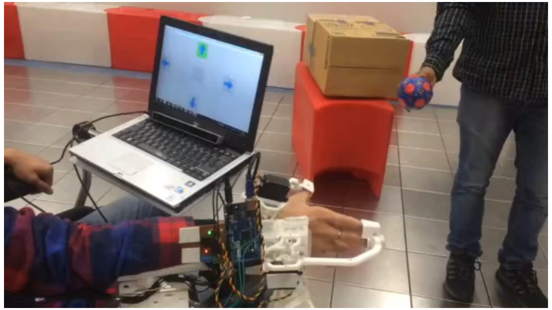
An integrated platform using dual panel-based GUI





(a) Wheelchair navigation panel

(b) Exoskeleton control panel



Meena, Y. K., et al.,. Emohex: An eye tracker-based mobility and hand exoskeleton device for assisting disabled people. In Proceedings of the IEEE-SMC.

Opportunities for Innovation

- Open-source accessibility tools
- Multilingual voice interfaces
- Offline-capable AI models
- Smartphone-based accessibility apps

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